



CodePulse

Intelligent Multi-Platform Coding Performance Analyzer

A unified dashboard that aggregates competitive-programming activity from Codeforces, LeetCode, AtCoder, and CodeChef — with AI-powered insights and recommendations.

Presented by

Abdus Salam Islam Badhon

BSSE ID: BSSE-1401 | Dept. of IIT, University of Dhaka

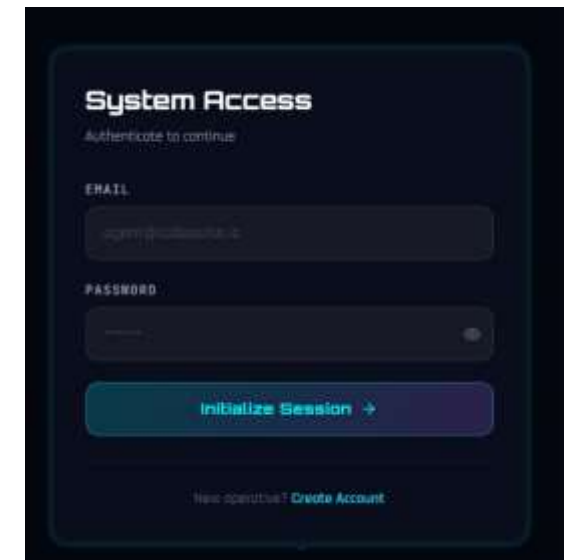
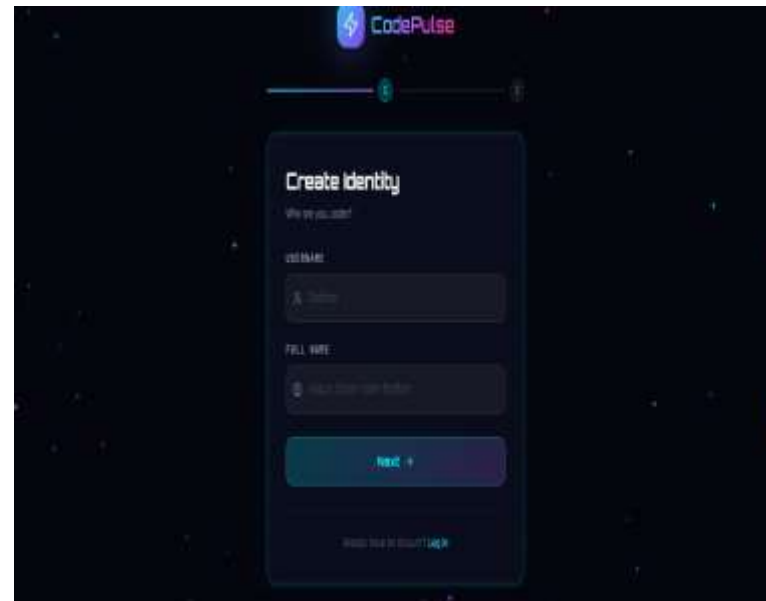
Supervised by

Mohd. Zulfiquar Hafiz

Professor, Dept. of IIT, University of Dhaka

Overview

- CodePulse is a full-stack web application that aggregates and analyses competitive-programming activity across Codeforces, LeetCode, AtCoder, and CodeChef into a single intelligent dashboard.
- Instead of checking four separate platforms, users get one unified view of their strengths, weaknesses, combined rating, and AI-generated problem recommendations.
- Built on a Next.js 14 frontend and a Spring Boot 3.2 backend, with Mistral AI powering the recommendation and coaching layer.



Motivation

1 Fragmented Data

Competitive programmers' stats are scattered across 4+ unrelated platforms with no shared view.

2 No Cross-Platform Insight

There is no easy way to see combined weaknesses, strengths, or a single unified rating.

3 Manual Problem Selection

Choosing what to practice next is largely manual, random, or based on gut feeling.

4 Limited Existing Tools

Existing trackers are single-platform and rarely offer AI-driven, personalised guidance.

5 Untapped Behavioural Data

Submission history hides patterns (streaks, blind spots, comfort zones) that go unused.

6 Opportunity for AI Coaching

Modern LLMs can turn raw submission history into a personal, context-aware coach.

Core Value Propositions

Problem (status quo)	CodePulse solution
Stats scattered across 4 platforms	Single combined dashboard
No cross-platform weakness detection	Unified topic weakness radar
Rating comparison between platforms is hard	CodePulse Rating — one number from all 4
Problem selection is manual / random	Algorithmic + AI-powered recommendations
No insight into behavioural patterns	Hidden Findings Engine (8+ auto-generated insights)
AI can't understand your coding history	Mistral AI receives your full performance context

What is CodePulse?

TARGET USERS

1 Developers

Track and improve competitive-programming performance across every platform they use.

2 Researchers

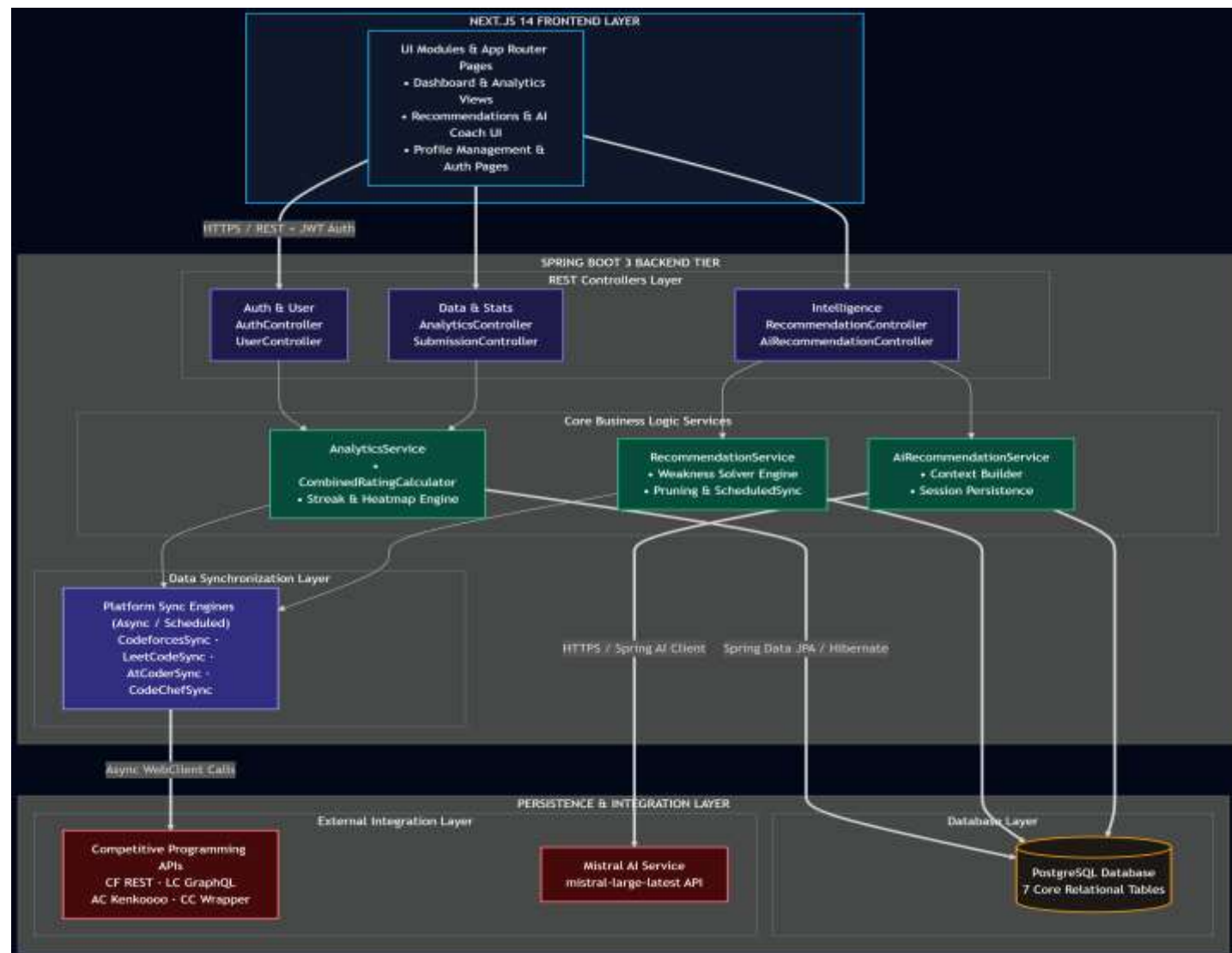
Study behavioural and performance patterns from aggregated multi-platform submission data.

3 Students / Teachers

Use unified analytics and AI recommendations for guided, structured practice.

System Architecture

Next.js frontend · Spring Boot backend · Data sync layer · PostgreSQL



Tech Stack

BACKEND

Layer	Technology
Language	Java 17
Framework	Spring Boot 3.2
Security	Spring Security + JWT + BCrypt
AI	Spring AI + Mistral (mistral-large-latest)
ORM	Spring Data JPA / Hibernate 6
Database	PostgreSQL 15+
Build	Maven 3.9

FRONTEND

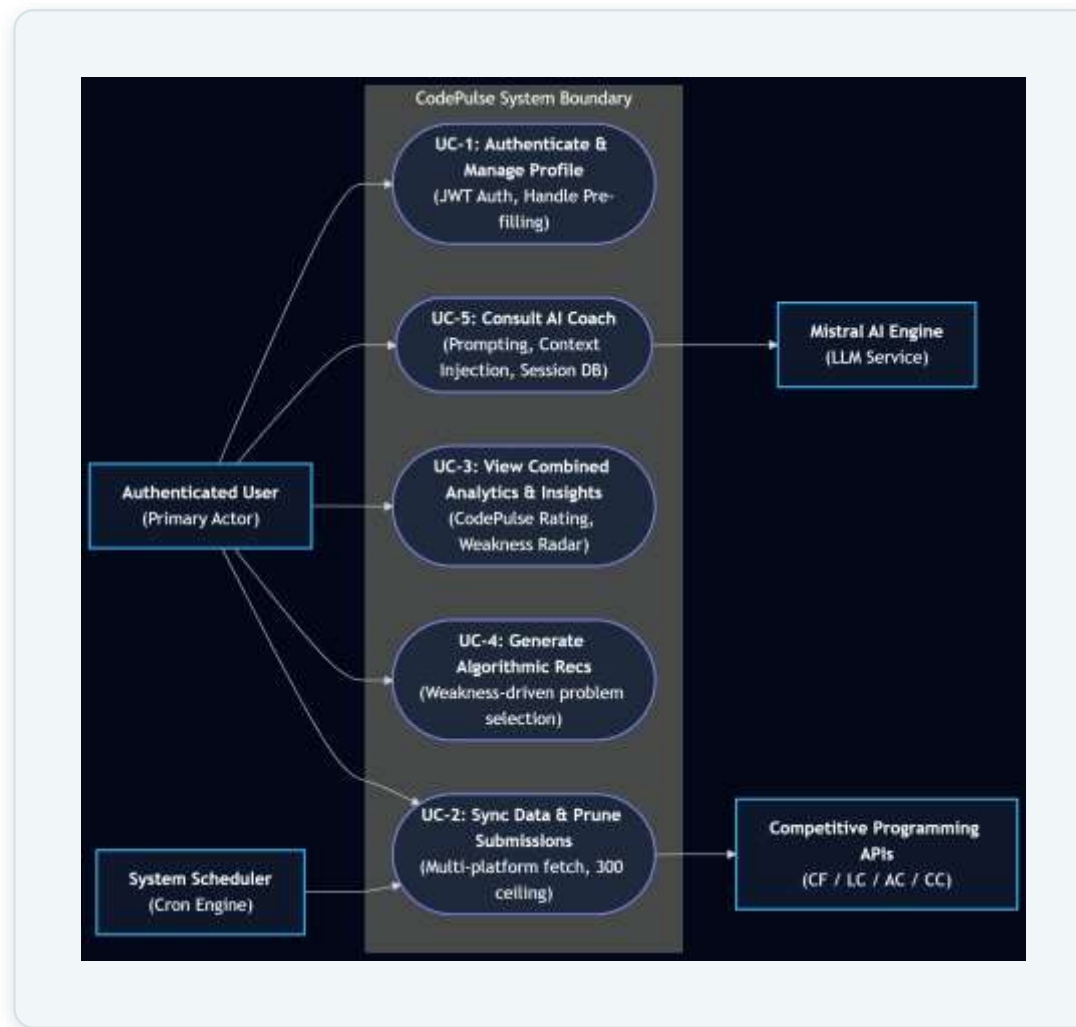
Layer	Technology
Framework	Next.js 14 (App Router)
Language	TypeScript
Styling	Tailwind CSS (cyberpunk theme)
Animation	Framer Motion
Charts	Recharts
State	Zustand + TanStack Query v5
HTTP	Axios

Supported Platforms

Platform	API	Data collected
Codeforces	Official REST API	Submissions, verdict, rating history, tags, difficulty
LeetCode	GraphQL API	Submissions, Easy/Medium/Hard difficulty, topic tags
AtCoder	kenkoooo.com community API	Submissions, Elo-like difficulty model, contest info
CodeChef	Community wrapper API	Solved problems, division-based difficulty estimation

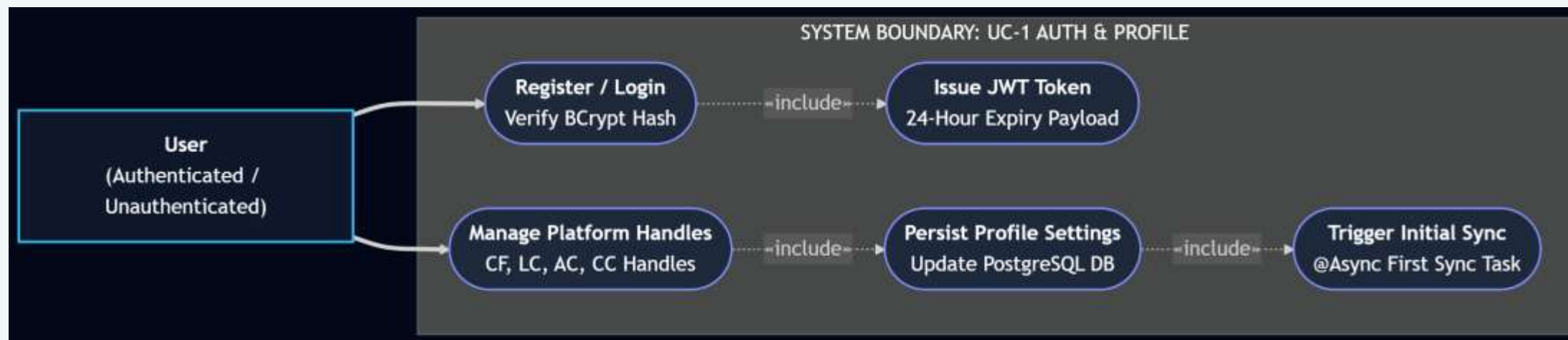
Use Case Diagram — System Overview

Five core use-case clusters (UC-1 to UC-5)



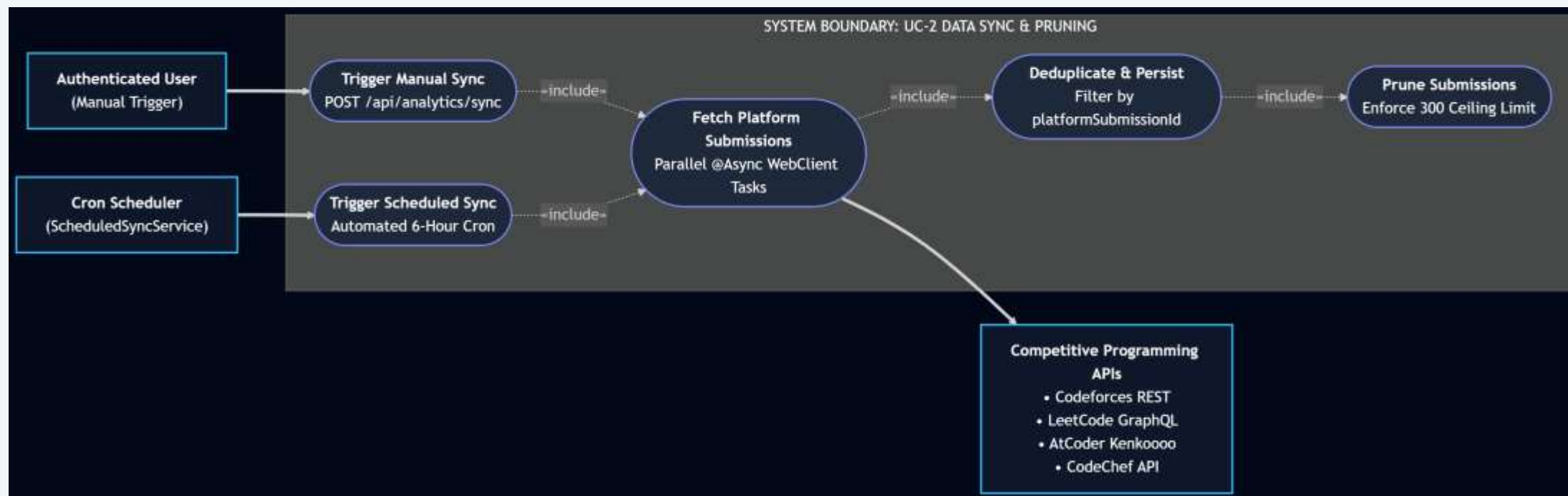
Use Case — Authentication & Profile

UC-1: registration, login, and platform-handle management



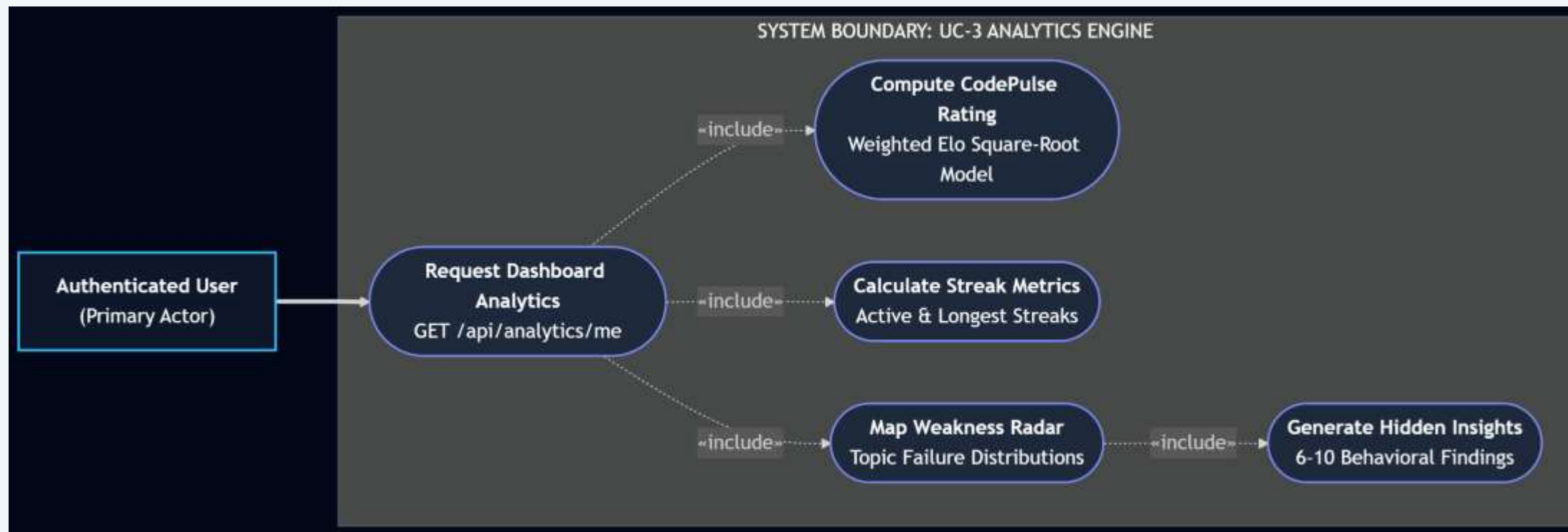
Use Case — Data Sync & Pruning

UC-2: multi-platform submission sync and retention



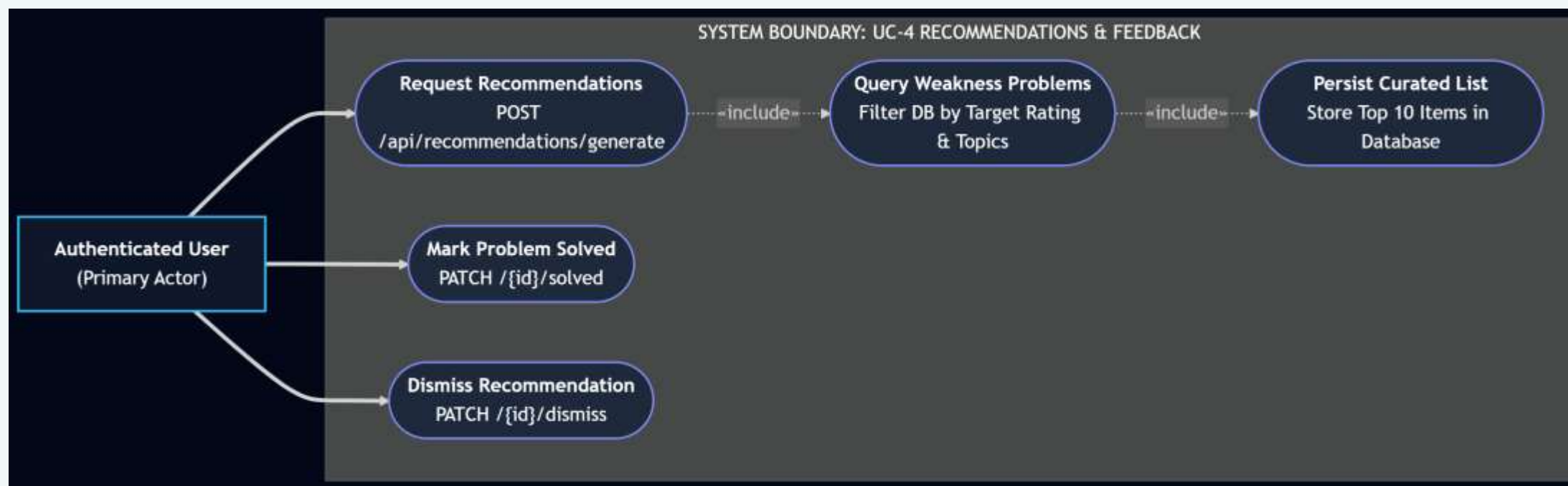
Use Case — Analytics Engine

UC-3: CodePulse Rating, streaks, and weakness radar



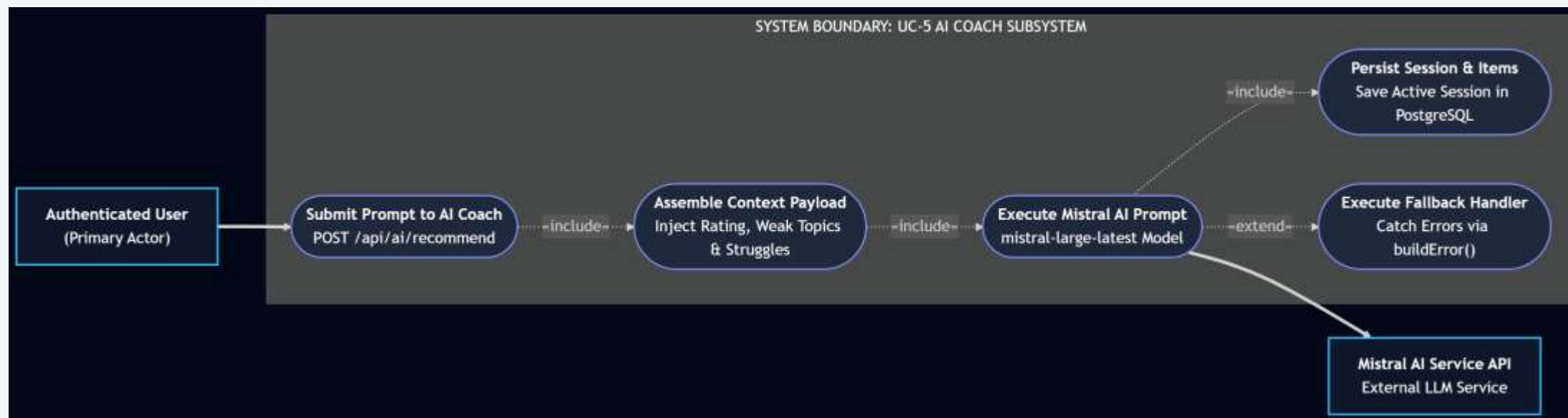
Use Case — Recommendations & Feedback

UC-4: curated problem generation and solve tracking



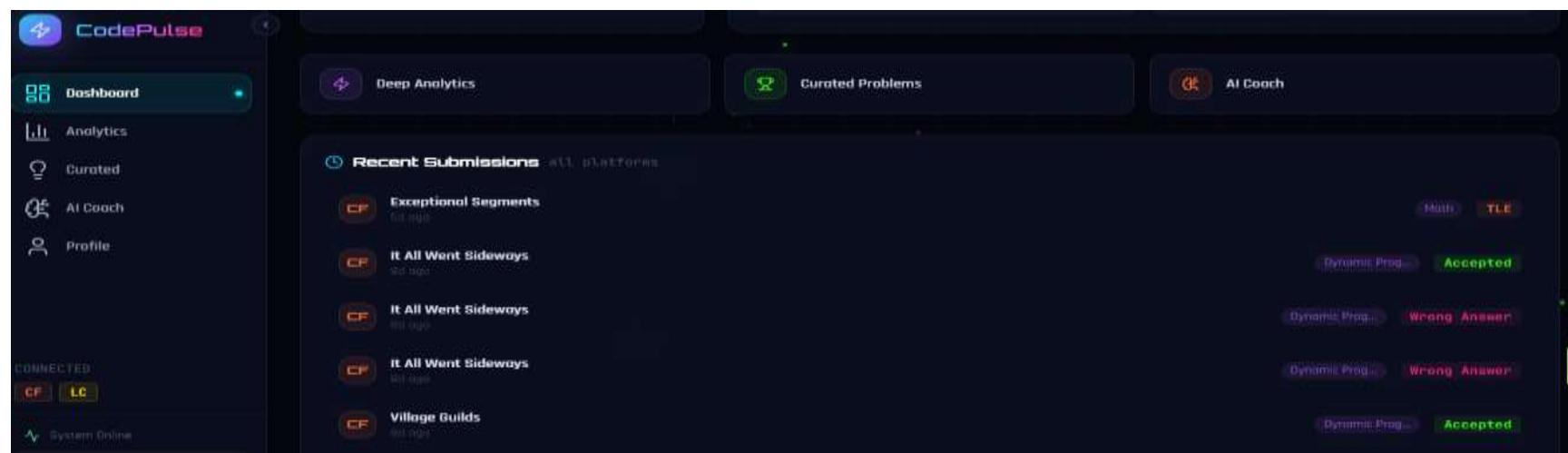
Use Case — AI Coach Subsystem

UC-5: prompt-driven Mistral AI recommendation sessions



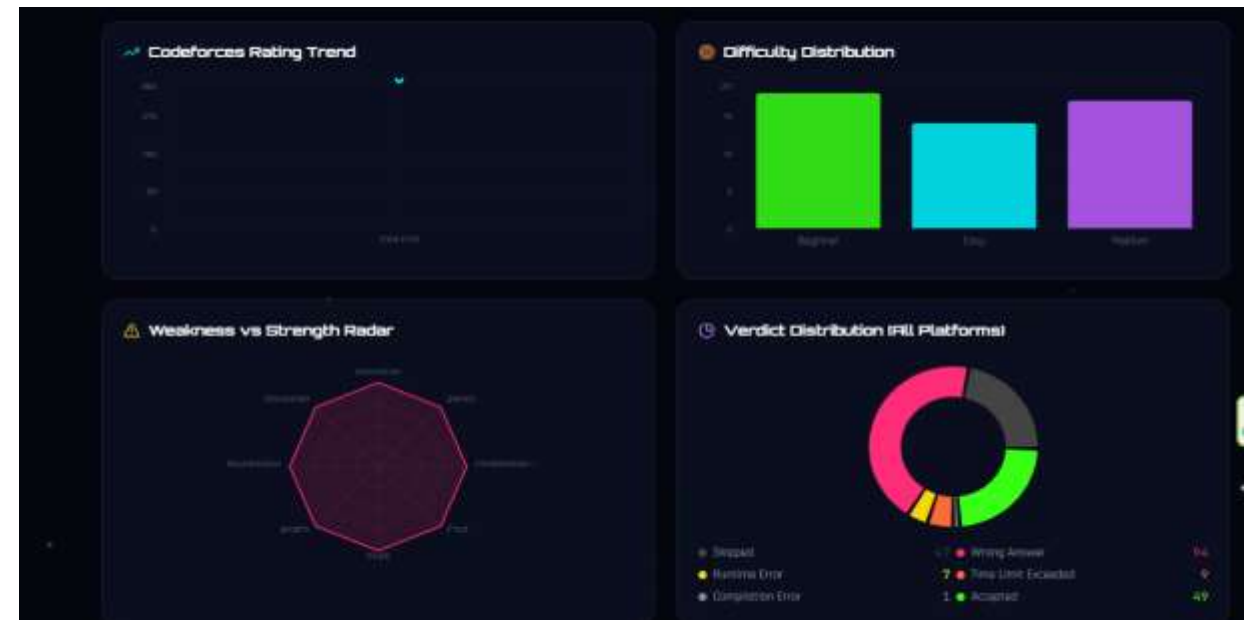
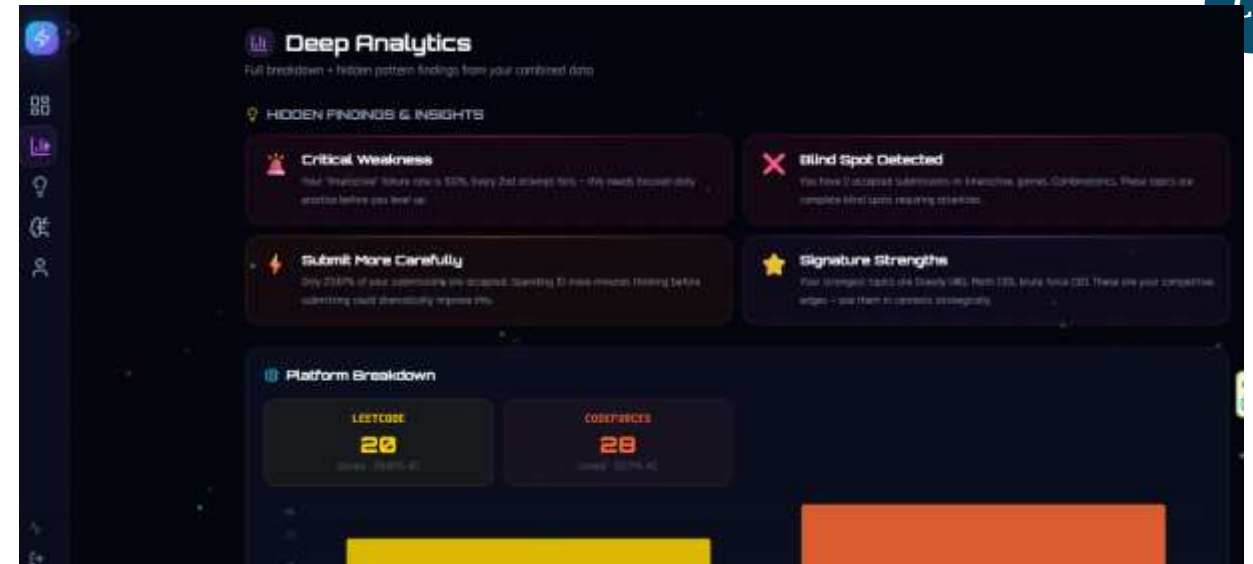
Feature — Combined Dashboard

- CodePulse Rating — unified rating computed from all 4 platforms
- Current & longest streak across all connected platforms
- Platform connection grid showing handle names and stats
- Recent submissions timeline with clickable links to the actual problems



Feature — Deep Analytics

- Activity heatmap — 26 weeks, all platforms combined
- Per-platform comparison chart & CF rating trend line
- Difficulty distribution normalised across platforms
- Topic coverage bar chart (top 12) & weakness vs strength radar
- Combined verdict distribution donut chart



Hidden Insights Engine

Auto-generates 6–10 findings from a user's combined data

1 Critical Weakness Detection

Flags the topic dragging performance down the most.

2 Zero-Accepted Topics

Surfaces blind-spot topics with no accepted submissions.

3 Acceptance Rate Analysis

Tracks accuracy trends across platforms.

4 Difficulty Comfort-Zone Alert

Warns when a user avoids harder problems.

5 Streak Pattern Recognition

Detects consistency and gaps in practice habits.

6 Topic Diversity Score

Measures breadth of topics practised.

7 Platform Imbalance Detection

Flags over-reliance on a single platform.

8 Activity Trend

Compares last 30 vs previous 30 days of activity.

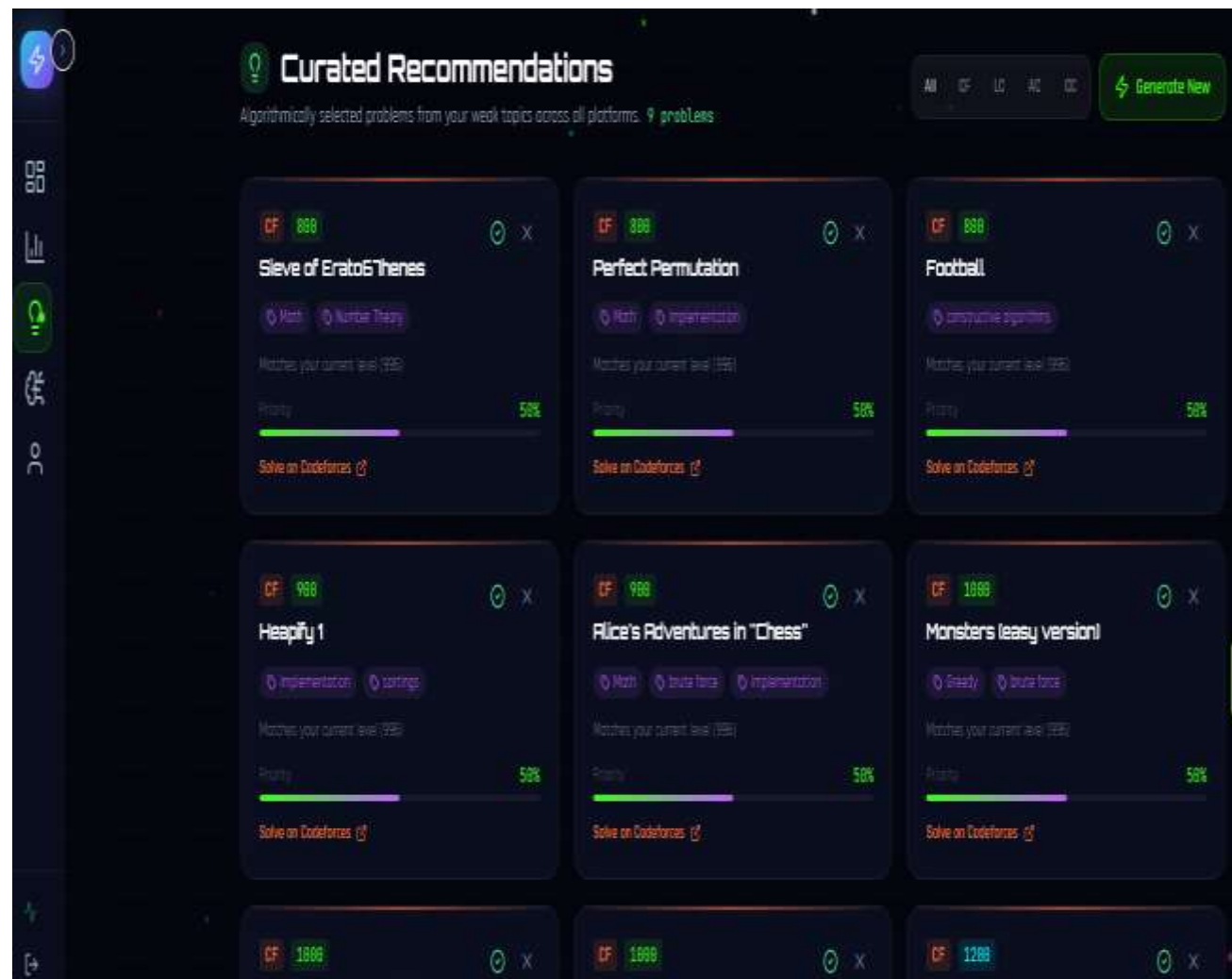
9 Strength Signature Recognition

Highlights the user's strongest topic areas.

Feature — Curated Recommendations

/recommendations

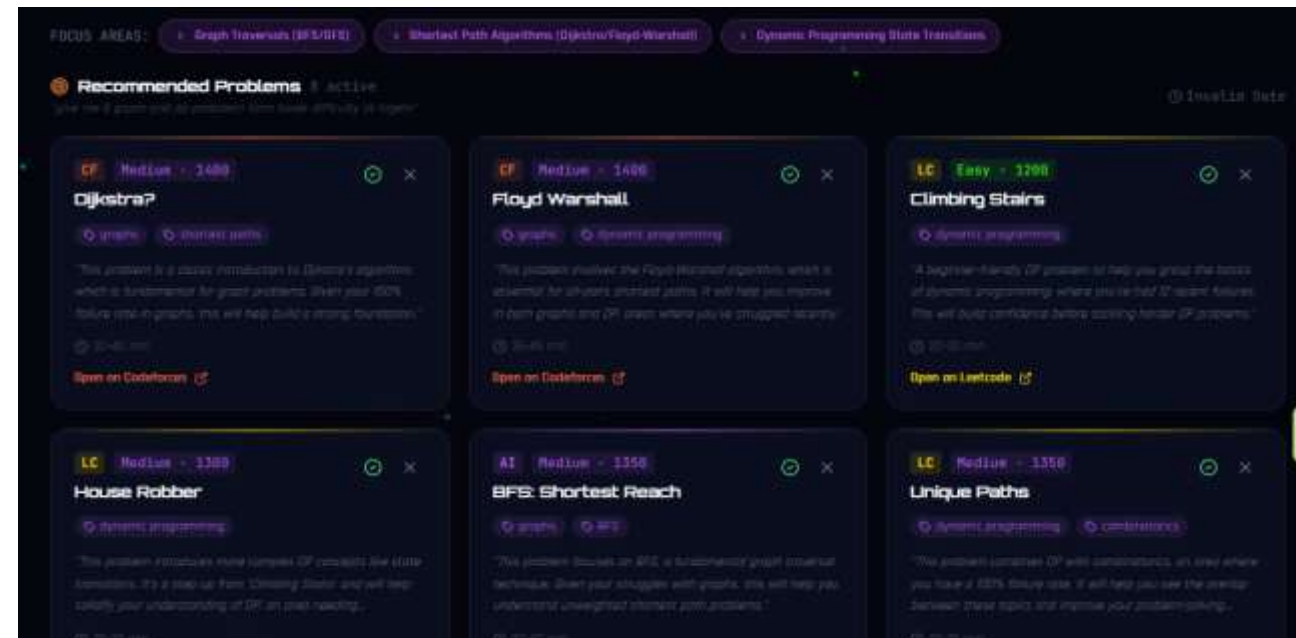
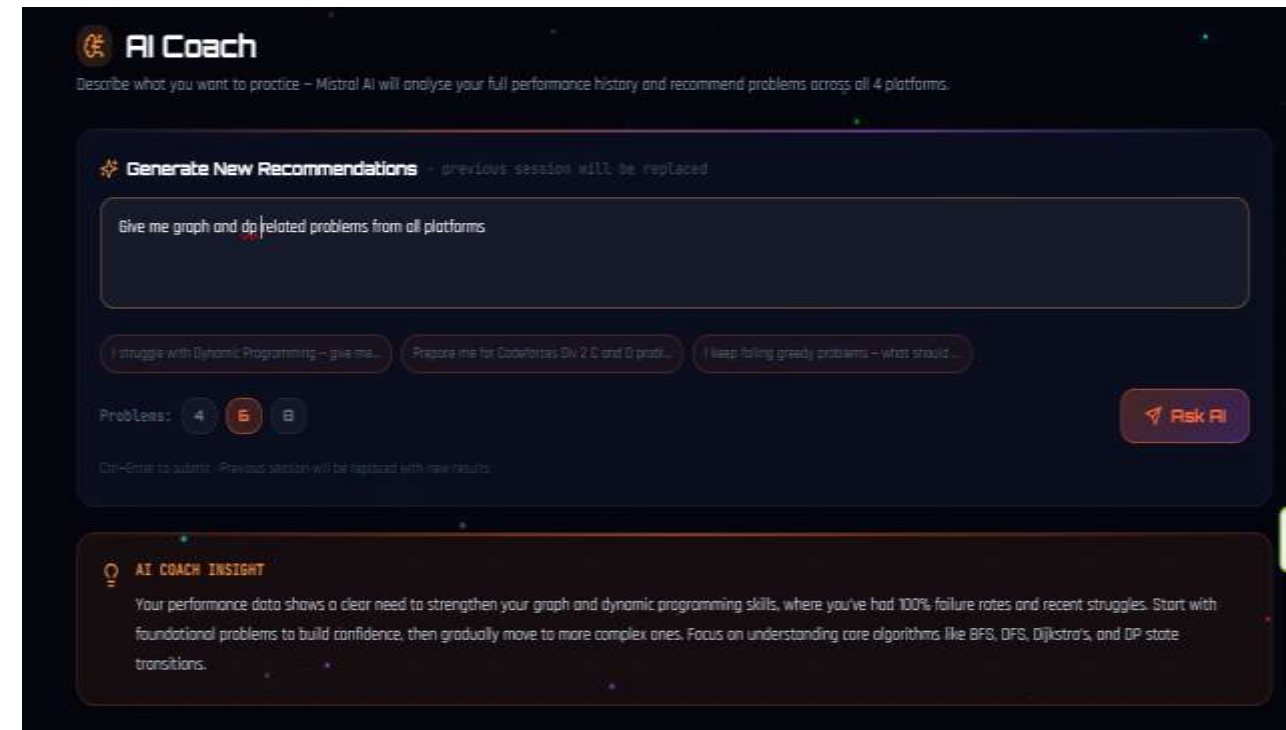
- Algorithmic recommendations based on detected weakness topics
- Estimated user rating determines the difficulty range
- Recommendations sourced from any of the 4 connected platforms
- Platform filter: All / CF / LC / AC / CC
- Mark solved (feedback loop) · Dismiss · Generate new (replaces previous)



Feature — AI Coach

[/ai-recommend](#)

- Free-text prompt, e.g. "I want to practice DP on trees"
- Mistral AI receives full user context: rating, weakness scores, per-platform stats, recently solved problems, recent WA/TLE topics
- Returns 4–10 personalised problem recommendations with direct URLs
- Coach Insight paragraph + Focus Areas
- Session stored in DB and shown on page load · Regenerate replaces previous session



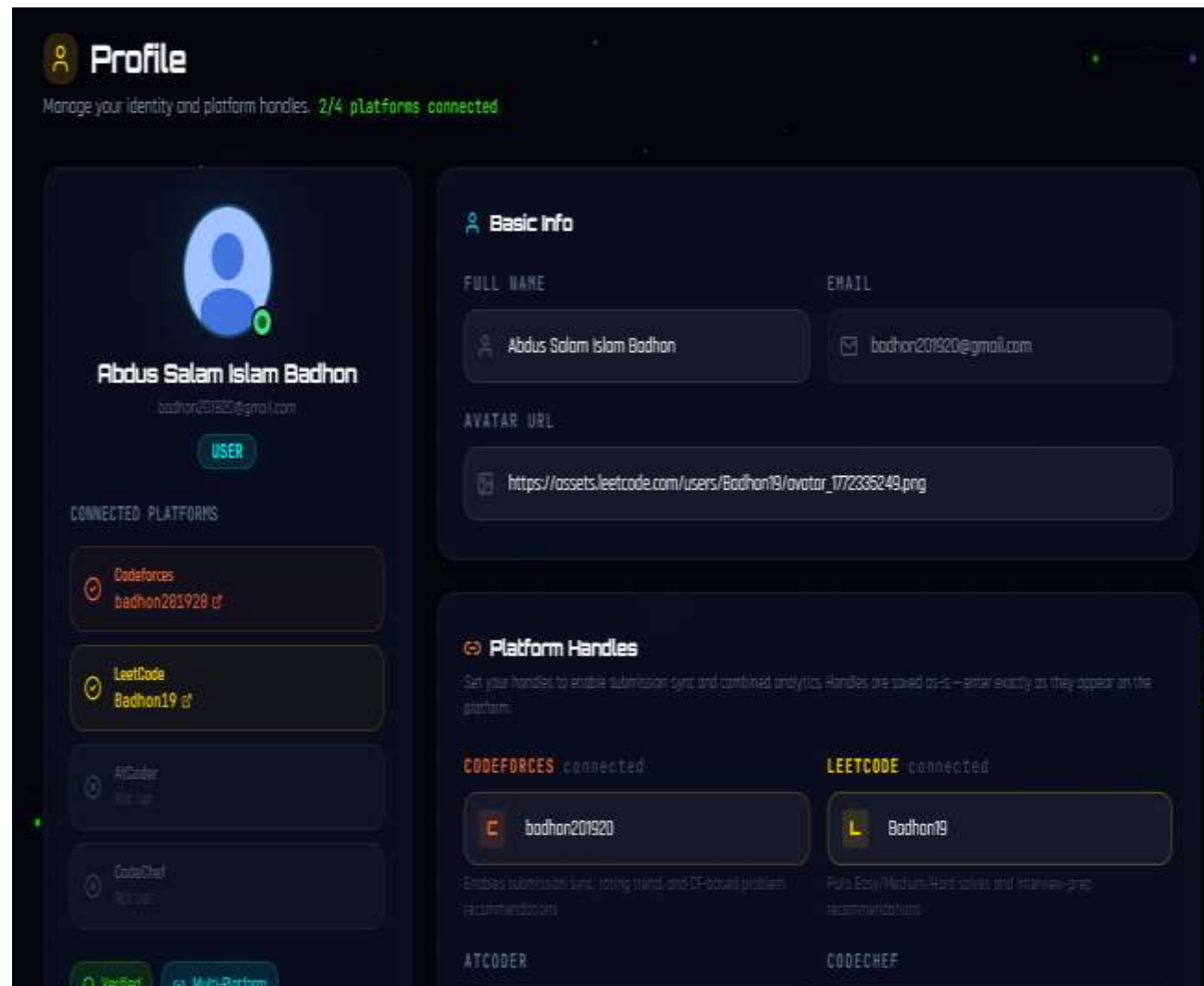
Feature — Profile & Auto-Sync

PROFILE

- All 4 platform handles always pre-filled from server
- Connection status badge per platform
- Clicking a handle opens the profile on that platform
- Saving triggers re-sync eligibility

AUTO-SYNC

- Runs every 6 hours for all active users (configurable cron)
- Triggered manually by the user from any page
- Submission pruning: keeps latest 300 per user (configurable)



CodePulse Rating Algorithm

A single Elo-like number representing combined performance

For each connected platform:

`CF_rating` = actual CF rating from /user.rating API (or estimated)

`LC_rating` = $1200 + (\text{easy} \times 2) + (\text{medium} \times 9) + (\text{hard} \times 22) + \text{acceptance_rate_bonus}$

`AC_rating` = average difficulty of solved AtCoder problems $\times 1.05$

`CC_rating` = $1000 + (\text{easy} \times 3) + (\text{medium} \times 10) + (\text{hard} \times 20) + (\text{expert} \times 35)$

Weight per platform = $\sqrt{(\text{submission_count_on_platform})}$

→ More activity on a platform gives it more influence, with diminishing returns

CodePulse_Rating = $\frac{\sum(\text{platform_rating} \times \text{weight})}{\sum(\text{weight})}$

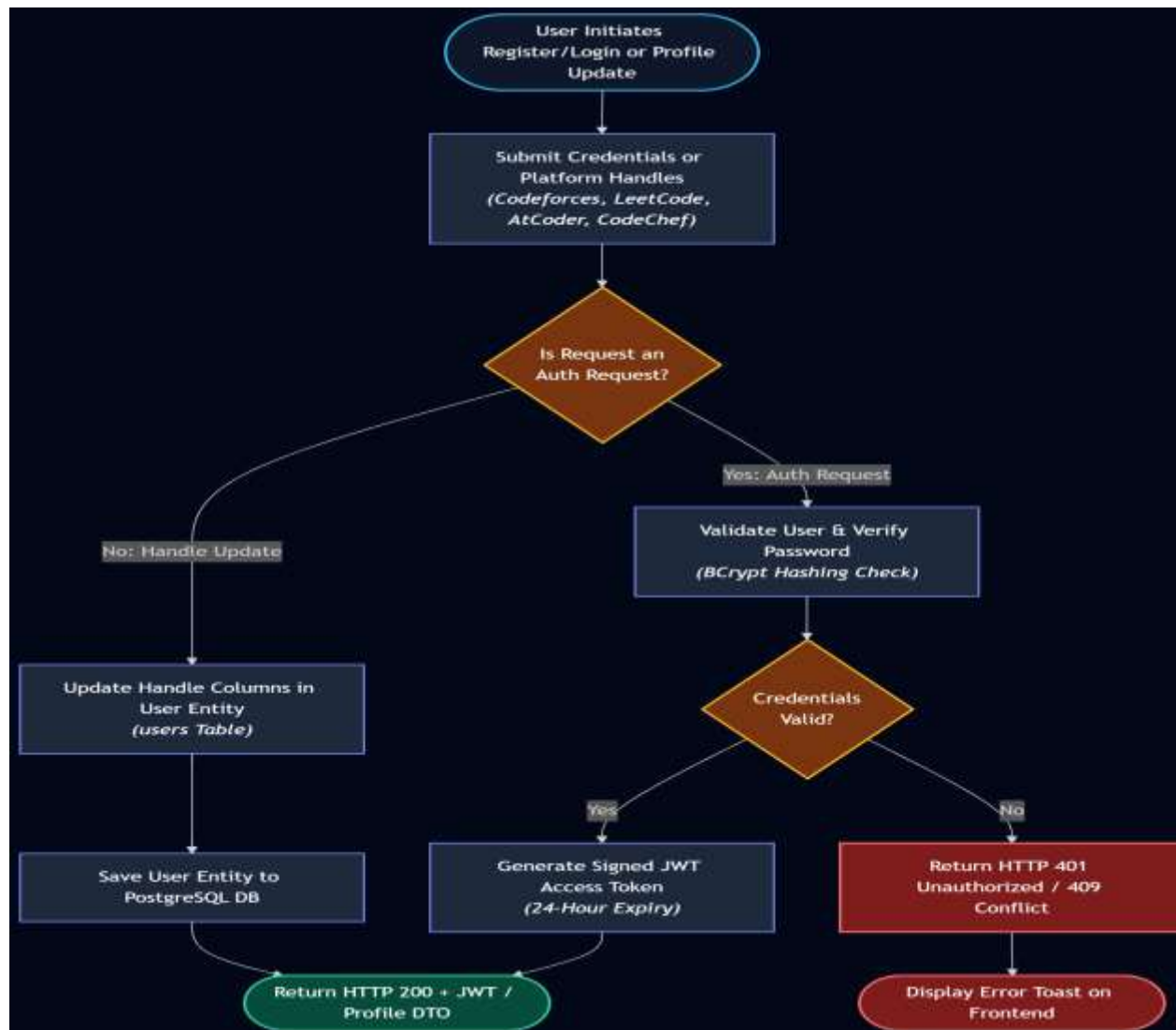
Rating Tier Mapping

Mirrors CF tier names for familiarity

Rating	Tier
< 1000	Beginner
1000 – 1199	Pupil
1200 – 1399	Apprentice
1400 – 1599	Specialist
1600 – 1899	Expert
1900 – 2099	Candidate Master
2100 – 2399	Master
2400 – 2599	International Master
2600 – 2999	Grandmaster
3000+	Legendary Grandmaster

Activity Diagram — Authentication & Profile

Register / login flow through JWT issuance and handle updates



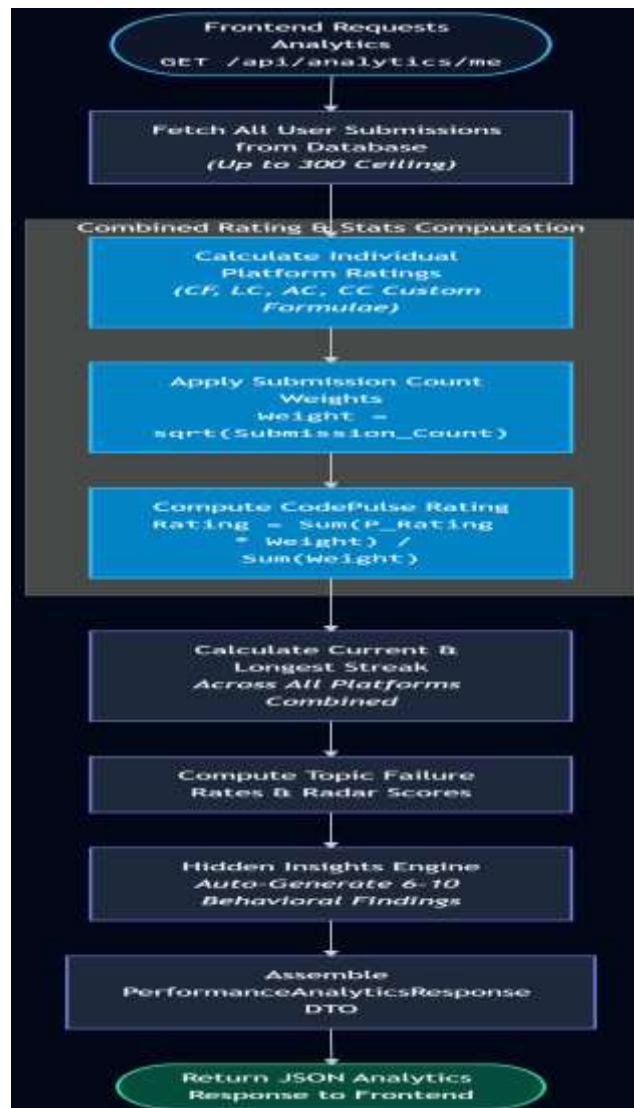
Activity Diagram — Data Sync & Pruning

Parallel multi-platform fetch → ingestion → pruning & finalisation



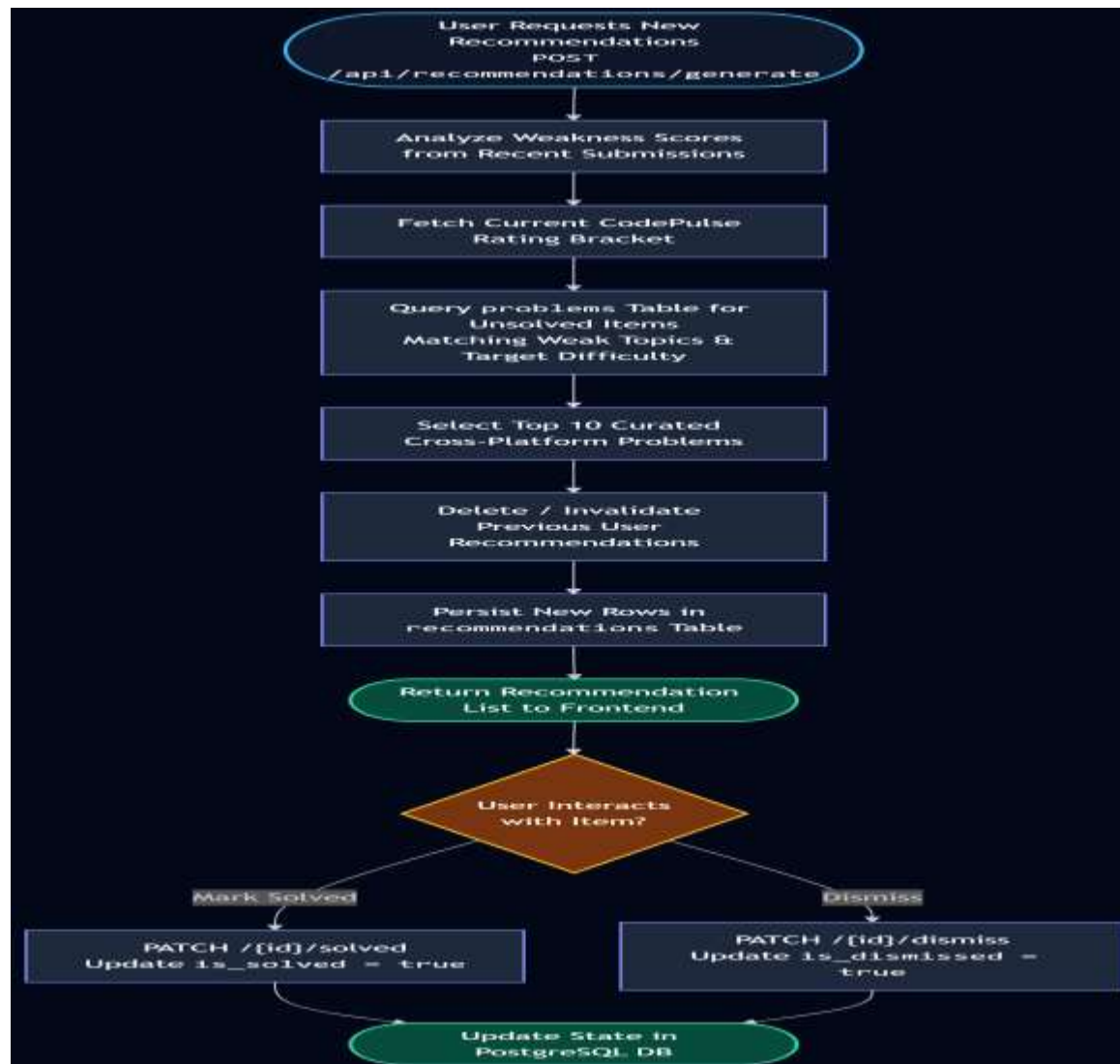
Activity Diagram — Analytics Computation

From GET /api/analytics/me to CodePulse Rating & hidden insights



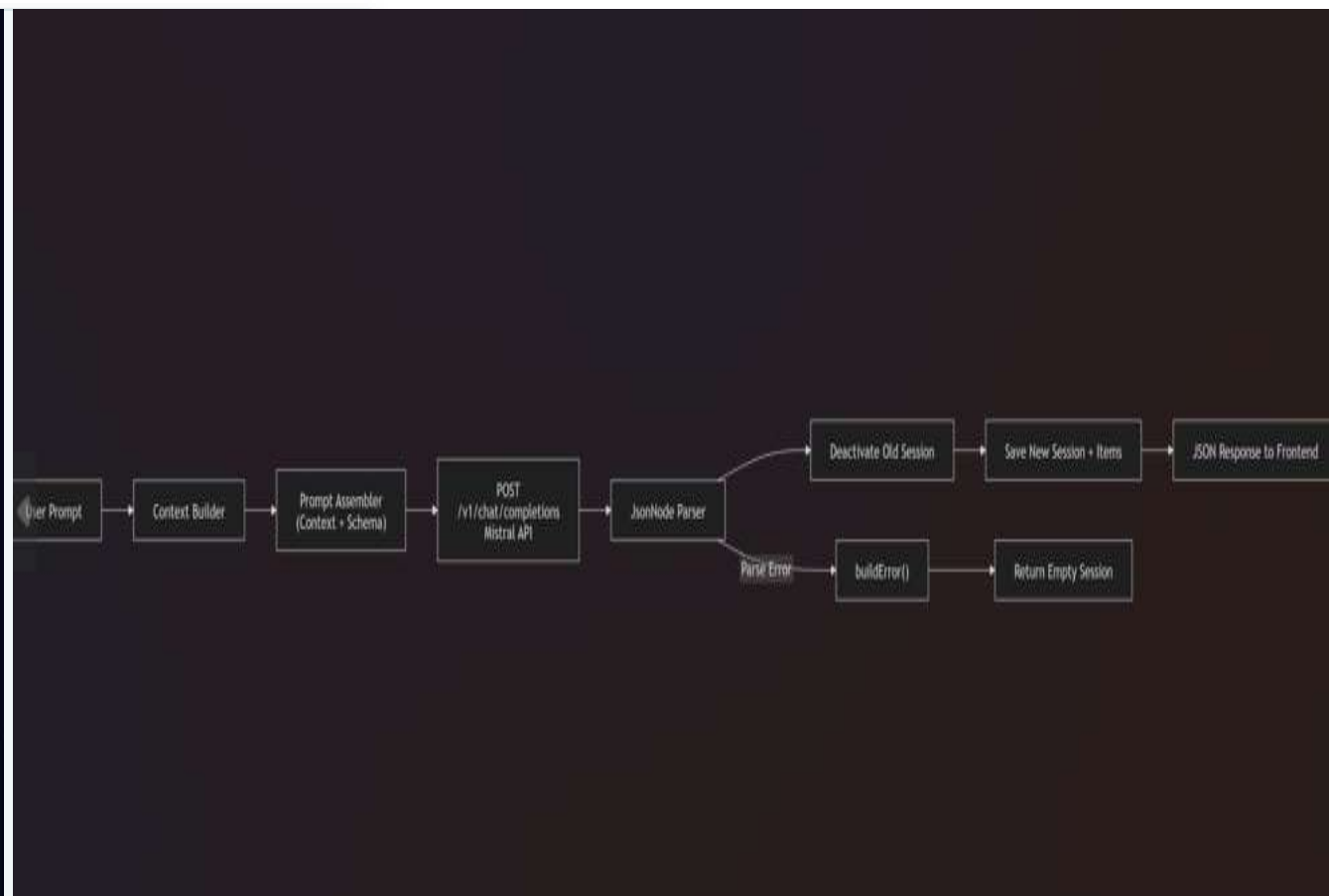
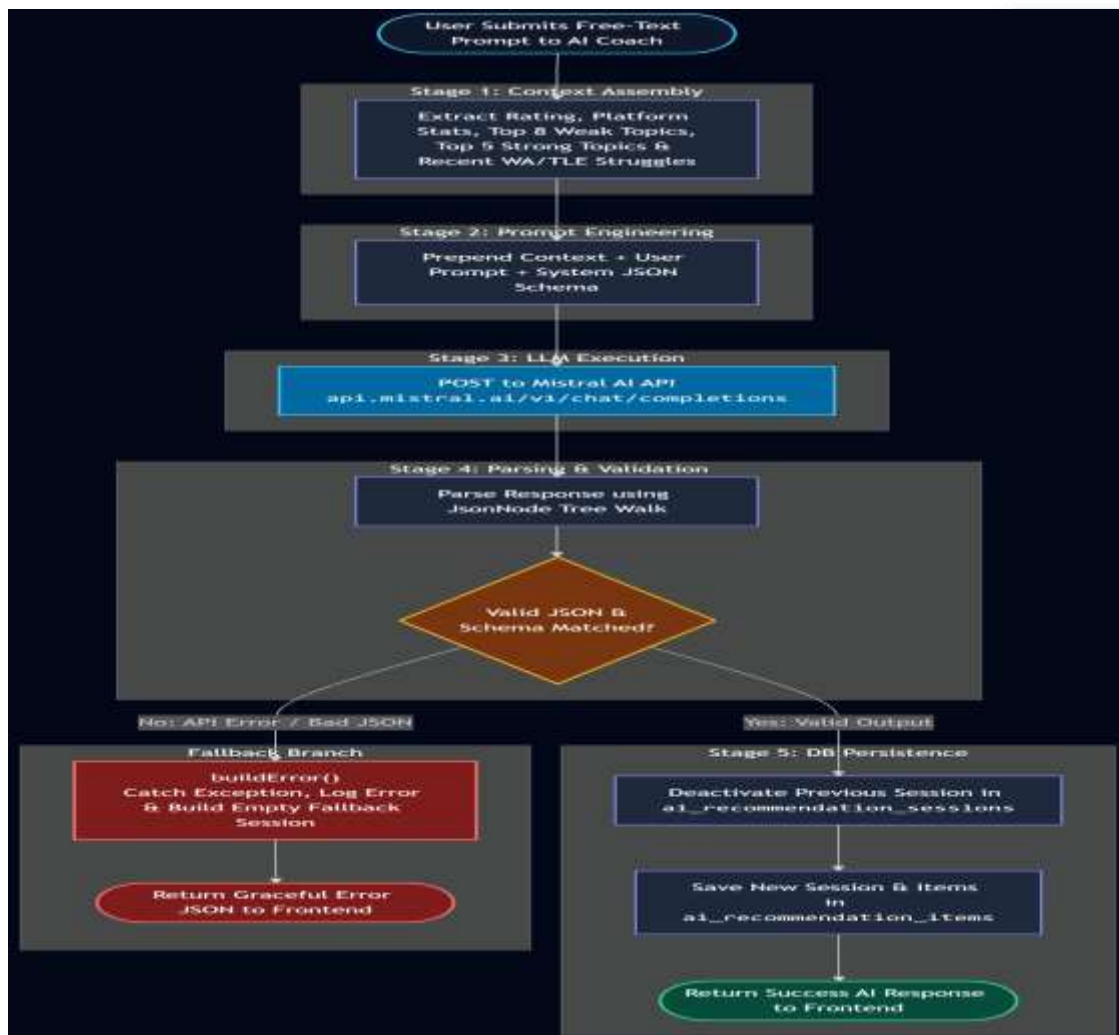
Activity Diagram — Recommendation Generation

Weakness analysis → problem matching → curated list persistence

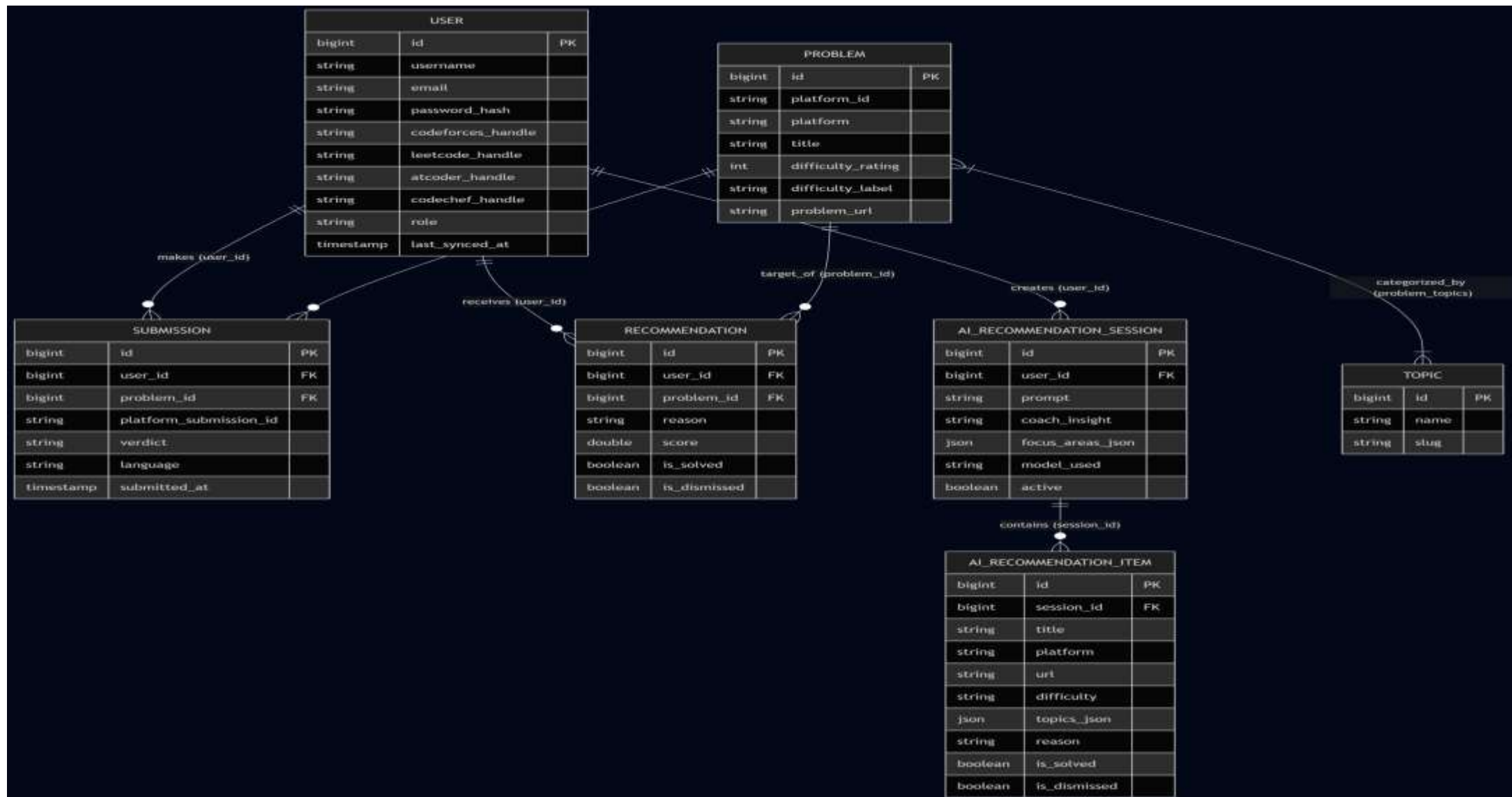


Activity Diagram — AI Coach Flow

Prompt → context assembly → Mistral AI call → parsing & validation



Entity-Relationship Diagram



Progress So Far

BACKEND

- Multi-platform data sync layer (CF, LC, AC, CC)
- CodePulse Rating algorithm implementation
- AI recommendation service integrated with Mistral
- JWT authentication & role-based access

FRONTEND

- Dashboard, Analytics & Recommendations pages
- AI Coach chat-style interface
- Login / Register flow with 2-step wizard
- Profile management with handle pre-fill

Timeline / Roadmap

Development phases, Weeks 1–16



References

- [1] Add citations for any papers, APIs, or libraries your project builds on.
- [2] Codeforces API documentation — <https://codeforces.com/apiHelp>
- [3] LeetCode GraphQL (community-documented) endpoints.
- [4] [kenkoooo.com](https://kenkoooo.com/atcoder/) — AtCoder Problems community API.
- [5] Mistral AI documentation — <https://docs.mistral.ai>
- [6] Spring AI reference documentation.

The background is a dark teal color. There are two large, lighter teal circular shapes: one in the top right corner and one in the bottom left corner, both partially cut off by the edges of the frame.

Thank You!

Questions & Feedback are welcome — CodePulse